

# **Extending COVID-19 Mortality Prediction into an Explainable Interactive Decision Support System**

Springboard Data Science Career Track  
Capstone Three Final Report

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## Executive Summary

This project extends the machine learning model developed in Capstone Two by transforming it into an interactive decision-support system emphasizing explainability, exploration, and user interaction. Rather than focusing solely on predictive accuracy, the project demonstrates how Explainable Artificial Intelligence (XAI), unsupervised learning, and interactive visualization can make machine learning models more transparent and useful for decision makers.

The Random Forest model developed previously serves as the analytical foundation for an interactive Streamlit dashboard that allows users to explore country-level COVID-19 data, modify predictor variables, generate updated mortality predictions, and understand those predictions through SHAP explanations. In addition, K-Means clustering is used to identify groups of countries with similar demographic, healthcare, and pandemic characteristics.

This report documents the design and implementation of the decision-support system, including the dashboard architecture, explainable AI components, clustering analysis, and interactive features that extend the original predictive model into a practical exploratory application.

## 1. Introduction

The predictive machine learning model developed in Capstone Two demonstrated that country-level demographic, healthcare, economic, vaccination, and policy variables could be used to predict cumulative COVID-19 mortality with meaningful accuracy. While predictive performance is an essential component of machine learning, many practical applications require more than accurate predictions alone. Users must also understand why a prediction was made, how changes to important variables influence outcomes, and how similar observations compare with one another.

Recent advances in Explainable Artificial Intelligence (XAI) have made it possible to interpret complex machine learning models without sacrificing predictive performance. Likewise, interactive visualization tools enable users to explore data, evaluate hypothetical scenarios, and better understand relationships that may not be apparent through static reports.

This project extends the Capstone Two Random Forest model by integrating SHAP-based model explanations, unsupervised country clustering, and an interactive Streamlit dashboard. The result is an explainable decision-support system that enables users to explore model behavior rather than simply viewing prediction results.

## 2. Project Objectives

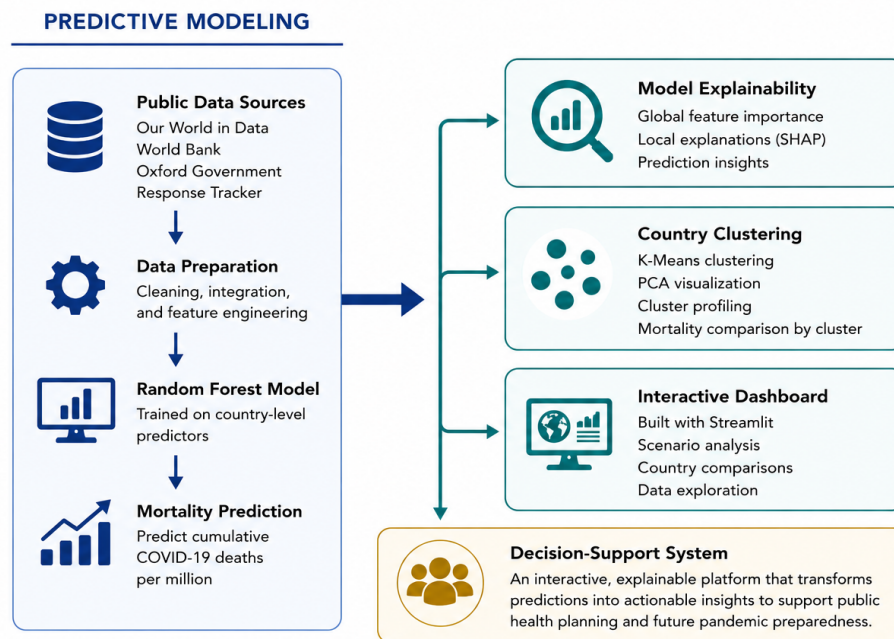
The primary objective of this project was to transform the predictive model developed in Capstone Two into an interactive, explainable decision-support application.

- Extend the previously developed Random Forest model into an interactive Streamlit dashboard.
- Implement SHAP explainability to provide both global and local model interpretations.
- Develop scenario analysis tools that allow users to modify country characteristics and observe resulting prediction changes.
- Apply K-Means clustering to identify groups of countries with similar demographic, healthcare, and pandemic characteristics.
- Provide interactive visualizations that support exploration of country-level COVID-19 data.
- Demonstrate how explainable AI can improve the transparency and usability of machine learning models.

## 3. System Architecture

The predictive model created during Capstone Two serves as the foundation for the decision-support system developed in this project. Rather than retraining the model, previously generated model artifacts—including the trained Random Forest model, preprocessing pipeline, and feature metadata—were incorporated into an interactive Streamlit application. Additional analytical components, including SHAP explainability and K-Means clustering, were integrated to provide greater insight into model behavior and country similarities.

**Figure 1, System Architecture of the Interactive COVID-19 Decision-Support Application.**



*This project develops a machine learning model to predict country-level COVID-19 mortality and extends the work with explainability, clustering, and an interactive dashboard to create a comprehensive decision-support system.*

The overall workflow begins with the curated country-level dataset and trained machine learning model. Predictions are generated dynamically as users modify input variables, while SHAP values provide feature-level explanations for each prediction. Clustering and visualization components allow users to explore broader patterns across countries, transforming the original predictive model into an interactive exploratory platform.

#### 4. Country Exploration Through Unsupervised Learning

Countries differ substantially in their demographic, healthcare, economic, and public health characteristics. Examining these variables individually provides valuable insight but may overlook broader patterns that emerge when countries are considered collectively. To support exploratory analysis within the decision-support system, an unsupervised learning approach was used to identify groups of countries with similar underlying characteristics.

K-Means clustering was applied to the country-level dataset developed in Capstone Two. Unlike supervised machine learning methods, which learn relationships between predictor variables and a known outcome, K-Means partitions observations into groups

based solely on similarity across the selected predictor variables. This approach provides a higher-level view of the data, allowing users to identify countries with comparable demographic, healthcare, economic, and public health profiles.

Prior to clustering, all predictor variables were standardized to ensure that variables measured on different scales contributed equally to the clustering solution. The analysis incorporated demographic, healthcare, economic, and public health indicators describing each country. Outcome variables, including COVID-19 mortality and other response variables, were intentionally excluded so that the clusters would reflect similarities in underlying country characteristics rather than similarities in outcomes. This provides an independent context for interpreting the predictions generated by the machine learning model.

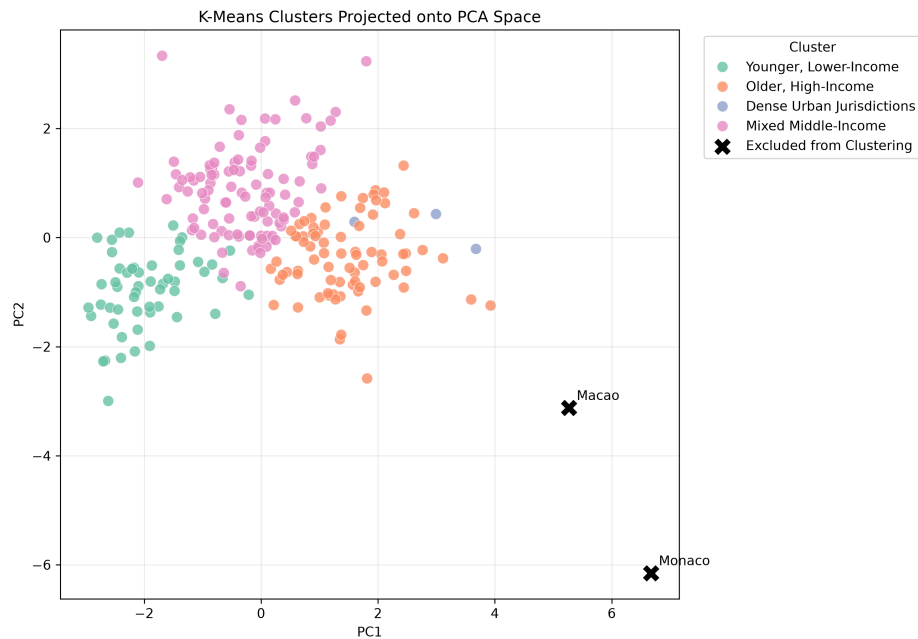
Before fitting the K-Means model, two countries—**Monaco and Morocco**—were removed from the clustering analysis because they exhibited unusually large distances from the remaining observations in the multidimensional feature space. These extreme observations distorted the cluster structure and reduced the interpretability of the resulting groups. Both countries remained in the overall dataset and Streamlit application but were excluded only from the clustering model used to develop the country groupings.

The optimal number of clusters was selected using the elbow method and silhouette analysis to balance model simplicity with the ability to identify meaningful groups of countries. Once the clustering solution was established, each country was assigned to a cluster and incorporated throughout the interactive dashboard.

Because clustering was performed in a multidimensional feature space, Principal Component Analysis (PCA) was used to project the country profiles onto two principal components for visualization. Although dimensionality reduction inevitably results in some loss of information, the PCA projection preserves the overall structure of the data and provides an intuitive representation of the relationships among the identified clusters.

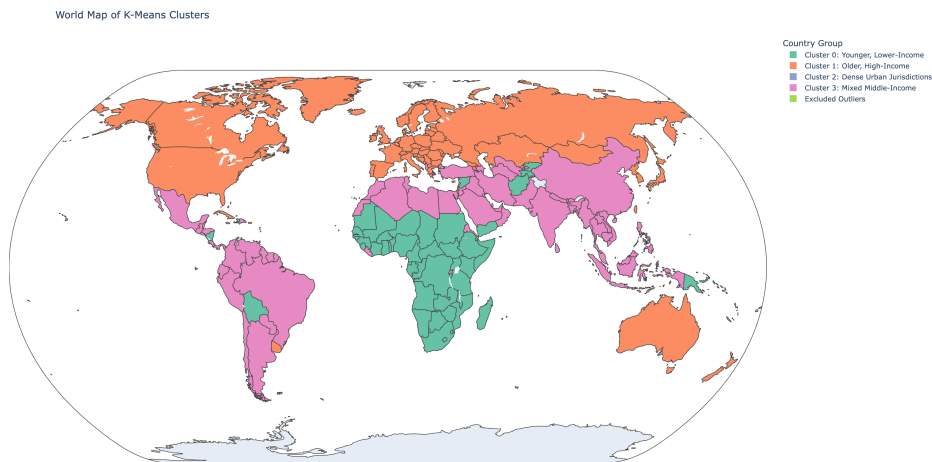
Rather than presenting clustering as a standalone analytical result, the cluster assignments were integrated throughout the Streamlit dashboard to support interactive exploration. Users can examine the geographic distribution of clusters, compare countries with similar underlying characteristics, and view mortality outcomes within the context of comparable nations. By combining unsupervised learning with interactive visualization, the dashboard provides an additional layer of context that complements the machine learning predictions and supports more informed interpretation of country-level outcomes.

**Figure 2. Principal Component Analysis of Country Clusters**



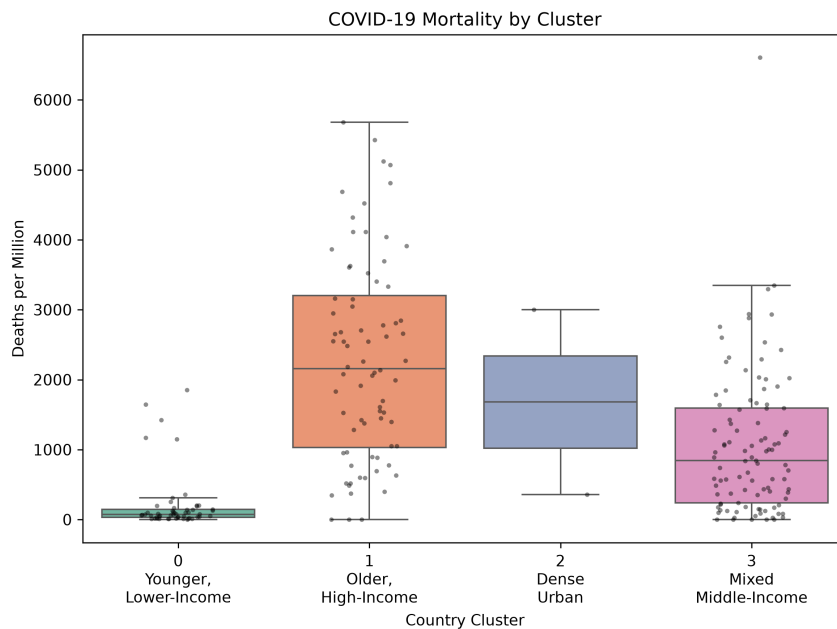
*The PCA projection summarizes the multidimensional clustering solution into two principal components, illustrating the overall separation of clusters while preserving the relative relationships among countries.*

**Figure 3. Interactive Global Map of Country Clusters'**



*Countries are colored according to their assigned K-Means cluster, allowing users to explore the geographic distribution of countries with similar demographic, healthcare, economic, and public health characteristics.*

**Figure 4. Distribution of COVID-19 Mortality by Cluster**



*Although COVID-19 mortality was not used to construct the clusters, the boxplots compare mortality distributions across the identified groups, demonstrating that countries with similar underlying characteristics also exhibit distinct mortality patterns.*

## 5. Interactive Decision-Support Dashboard

The primary objective of Capstone Three was to transform the predictive machine learning model developed in Capstone Two into an interactive decision-support application. While the original project demonstrated that Random Forest models could accurately predict COVID-19 mortality, the results were presented primarily through static analyses. The Streamlit dashboard extends this work by allowing users to explore country characteristics, generate real-time predictions, and interpret model behavior through an intuitive web-based interface.

The dashboard was developed using the Streamlit framework, which enables rapid deployment of data science applications through a browser-based interface. The application integrates the trained Random Forest model, country clustering results, interactive visualizations, and explainable artificial intelligence (XAI) techniques into a unified analytical environment. This integration allows users to move beyond static reports and actively explore relationships within the data.

The application is organized into a series of interactive pages designed to support different aspects of the analytical workflow. Users begin by exploring country characteristics and cluster assignments before progressing to scenario analysis and prediction. Interactive controls allow users to modify predictor values and immediately observe the corresponding changes in predicted mortality. This capability enables

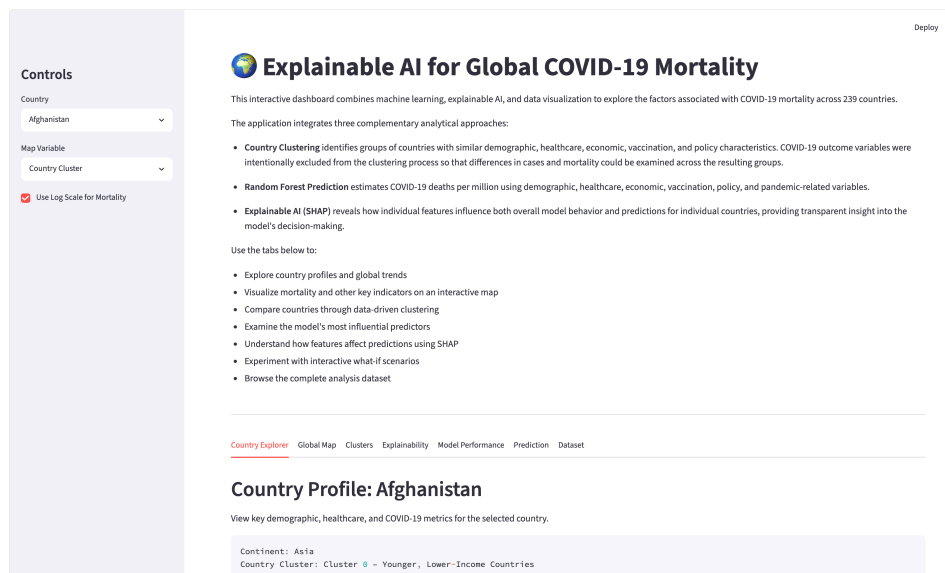
exploration of hypothetical scenarios and provides insight into how changes in demographic, healthcare, or public health indicators influence model predictions.

In addition to prediction, the dashboard incorporates visualizations that summarize the underlying dataset, cluster membership, and model outputs. Interactive maps, comparison charts, and summary statistics provide multiple perspectives on the data while allowing users to focus on countries or regions of interest. These visualizations complement the predictive model by providing contextual information that supports interpretation of the results.

A central feature of the dashboard is its integration of explainable artificial intelligence. Rather than presenting predictions as isolated numerical outputs, the application enables users to investigate the factors contributing to each prediction using SHAP (SHapley Additive exPlanations). This functionality provides transparency into the Random Forest model and supports more informed interpretation of the prediction results. The explainability component is discussed in detail in the following section.

By combining predictive modeling, unsupervised learning, interactive visualization, and model explainability within a single application, the Streamlit dashboard transforms a traditional machine learning model into a practical decision-support system. The resulting application provides an accessible platform for exploring complex relationships among country characteristics while improving the transparency and interpretability of machine learning predictions.

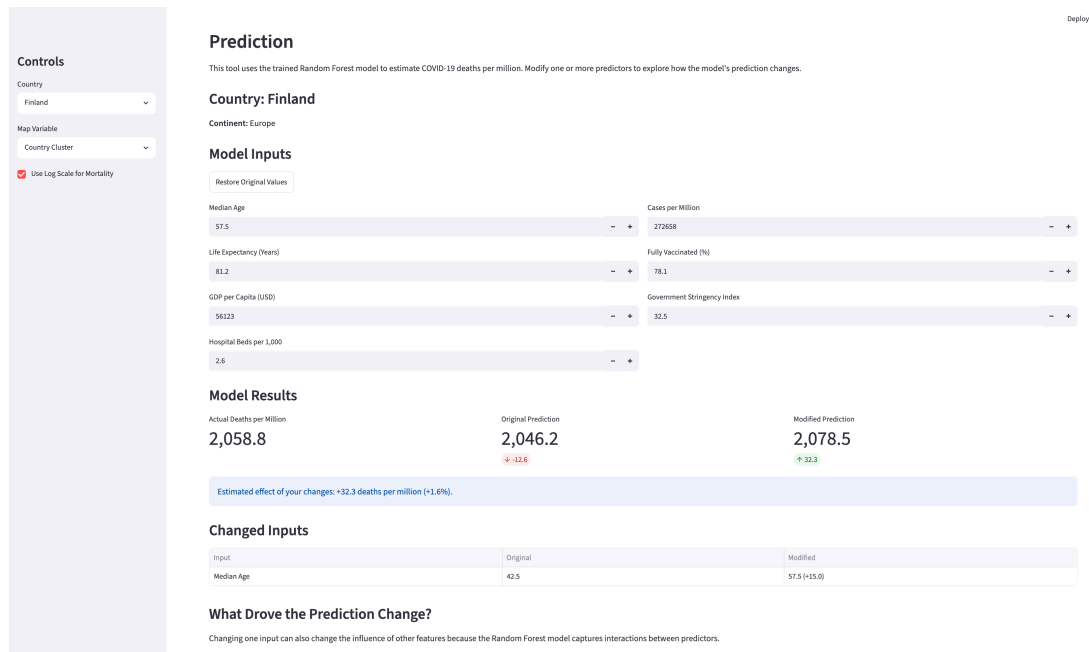
## Figure 5. Streamlit Dashboard Overview



*The main dashboard provides users with access to interactive visualizations, country exploration tools, clustering results, predictive modeling, and explainable AI components through a unified web-based interface.*



**Figure 6. Interactive Prediction and Scenario Analysis**



*Users can modify predictor variables to evaluate hypothetical scenarios and immediately observe changes in the predicted COVID-19 mortality rate, enabling real-time exploration of model behavior and decision support.*

## 6. Explainable Artificial Intelligence

While predictive accuracy is an important measure of machine learning performance, accurate predictions alone provide limited value if users cannot understand how those predictions are generated. Ensemble methods such as Random Forest are often considered "black-box" models because their internal decision-making processes are difficult to interpret directly. To improve transparency and support informed decision making, the Streamlit application incorporates Explainable Artificial Intelligence (XAI) through SHapley Additive exPlanations (SHAP).

SHAP is a model-agnostic explainability technique based on cooperative game theory that assigns each predictor a contribution value representing its influence on a model prediction. Positive SHAP values increase the predicted mortality, while negative values decrease it. Unlike traditional feature importance measures, SHAP provides both global explanations that summarize model behavior across the entire dataset and local explanations that describe the factors influencing an individual prediction.

Global SHAP analysis was used to identify the variables that consistently exert the greatest influence on the Random Forest model. Figure 7 ranks the predictor variables according to their average absolute SHAP values, providing an overall measure of each feature's contribution to model predictions. The results demonstrate that the model relies

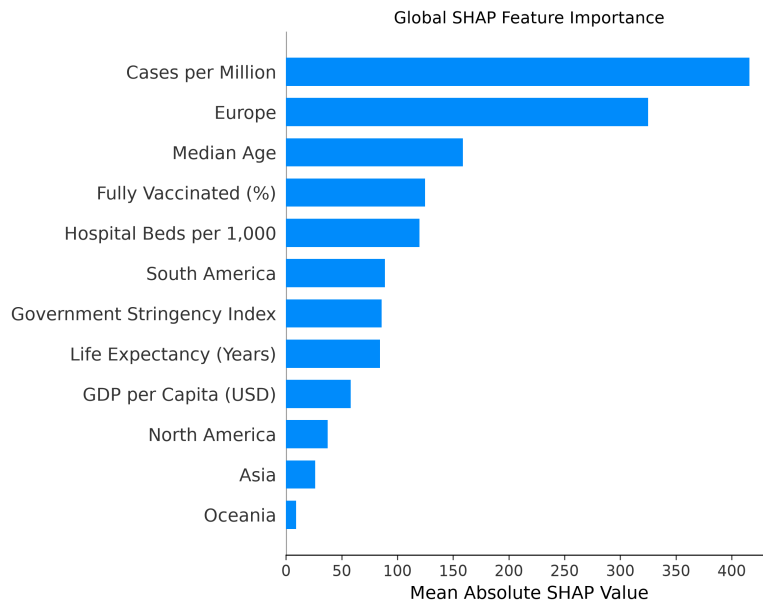
on a combination of demographic, healthcare, vaccination, and public health characteristics rather than a single dominant predictor, reflecting the multifactorial nature of COVID-19 mortality.

The interactive dashboard extends traditional SHAP analysis by integrating explainability directly into the prediction workflow. After users modify one or more predictor variables, the application immediately recalculates both the predicted mortality and the associated SHAP values. Rather than simply displaying a revised prediction, the dashboard compares SHAP contributions before and after the changes to identify how the model's reasoning has evolved.

To improve interpretation, the application distinguishes between **direct effects** and **model interactions**. Direct effects represent the influence of predictor variables that were intentionally modified by the user, while model interactions identify variables whose contributions changed even though they were not directly edited. Because the Random Forest model captures complex relationships among predictors, changing a single variable may alter the influence of several others. The dashboard summarizes these indirect effects separately, allowing users to better understand the interconnected nature of the model.

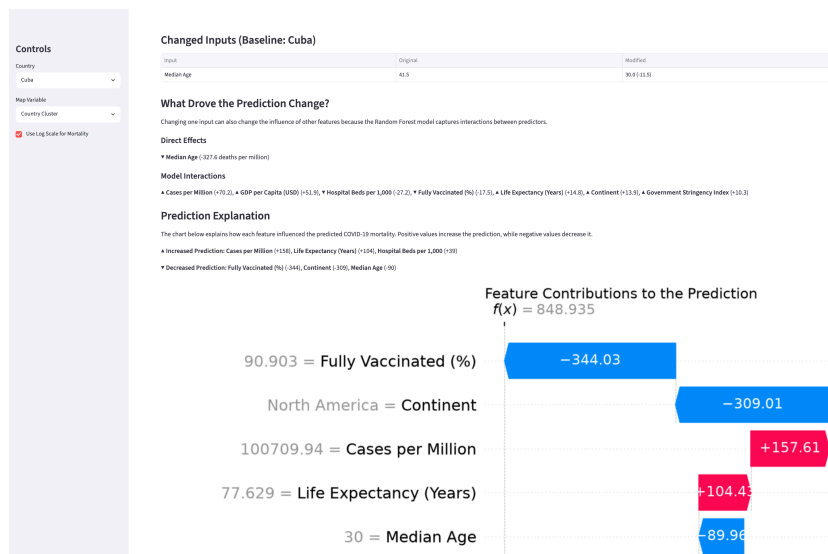
Figure 8 illustrates the integrated prediction explanation interface within the Streamlit application. Following a user-defined scenario, the dashboard presents the revised prediction, summarizes the modified inputs, identifies the direct and indirect contributors to the change in prediction, and visualizes the final prediction using a SHAP waterfall plot. By combining prediction, scenario analysis, and explainability within a single interface, the application transforms machine learning from a static predictive model into an interactive decision-support system that enables users to both generate and understand predictions.

**Figure 7. Global SHAP Feature Importance**



*Global SHAP values rank predictor variables according to their overall contribution to the Random Forest model, providing a summary of the factors that most strongly influence mortality predictions across all countries.*

**Figure 8. Interactive Prediction Explanation**



*The dashboard explains a user-defined prediction scenario by identifying modified inputs, distinguishing direct effects from indirect model interactions, and visualizing feature contributions using SHAP. The explanation updates dynamically as users adjust model inputs, providing transparency into both the prediction and the model's underlying reasoning.*

Absolutely. Given everything we've developed over the last couple of days, I'd end the report with a single **Conclusion** section rather than separating Discussion and Conclusion. It keeps the report concise and leaves the reader with a strong summary of what was accomplished.

## 7. Conclusion

This project developed an interactive decision-support application for exploring the factors associated with country-level COVID-19 mortality. Building upon the predictive modeling developed in Capstone Two, Capstone Three extended the project by integrating unsupervised learning, interactive visualization, and explainable artificial intelligence into a unified Streamlit application. Rather than presenting machine learning results as static analyses, the application enables users to explore relationships within the data, generate hypothetical scenarios, and interpret model predictions through an accessible web-based interface.

The Random Forest regression model provides mortality predictions using demographic, healthcare, vaccination, and pandemic-related characteristics. To complement these predictions, K-Means clustering groups countries with similar underlying characteristics, allowing users to compare nations beyond geographic location or mortality outcomes alone. By intentionally excluding mortality from the clustering process, the application identifies meaningful similarities based solely on country characteristics, providing additional context for interpreting model predictions.

The Streamlit dashboard transforms these analytical models into an interactive decision-support environment. Users can explore country clusters, visualize global patterns, modify predictor variables to evaluate hypothetical scenarios, and immediately observe the resulting changes in predicted mortality. This interactive workflow allows users to investigate how changes in individual characteristics influence model predictions while encouraging exploration of relationships that would be difficult to identify through static visualizations alone.

To improve transparency, the application integrates SHAP explainability directly into the prediction workflow. Global SHAP analysis identifies the variables that most strongly influence predictions across all countries, while local SHAP explanations reveal the factors driving individual predictions. By comparing SHAP values before and after user modifications, the application further distinguishes direct effects from indirect model interactions, allowing users to understand not only how predictions change but also why they change. This integration transforms explainability from a static analytical technique into an interactive component of the decision-support process.

Although the application demonstrates the value of combining machine learning with interactive analytics, several limitations should be acknowledged. The analysis relies on country-level aggregated data and therefore does not capture within-country variation or temporal changes over the course of the pandemic. Additionally, the predictive model

identifies statistical associations rather than causal relationships, and its performance is dependent on the completeness and quality of the publicly available data used for training.

Future work could expand the application by incorporating longitudinal data to model changes over time, evaluating additional machine learning algorithms, integrating more recent public health datasets, and allowing users to upload custom datasets for analysis. Additional scenario analysis features and expanded policy variables could further enhance the application's usefulness as an exploratory decision-support tool.

Overall, this project demonstrates how predictive modeling, unsupervised learning, interactive visualization, and explainable artificial intelligence can be integrated into a cohesive analytical framework. By combining these techniques within a single interactive application, the project moves beyond traditional predictive modeling to provide a transparent and accessible platform for exploring the complex factors associated with COVID-19 mortality across countries.